

SIMATS ENGINEERING

ASSIGNMENT

COURSE: MACHINE LEARNING (ITA 0603)

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Unit – Data Analysis with Pandas: DataFrame Operations, Grouping, Aggregation, Statistics, Filtering, Sorting, Plotting

Compare Income Distributions Among Different Regions Using Pandas and Statistical Measures

1. Problem Understanding

Objective:

To analyze and compare household/individual income distributions across different geographic regions using Pandas and standard statistical measures, in order to quantify the degree of income inequality within and between regions and highlight areas that most need policy attention.

Problem Statement:

Given a dataset of income records tagged by region (e.g., state, district, or urban/rural classification), compare how income is distributed across these regions using descriptive statistics (mean, median, standard deviation, skewness), inequality measures (Gini coefficient, percentile ratios), and statistical hypothesis testing (ANOVA / Kruskal-Wallis) to determine whether the observed regional differences in income are statistically significant, and to identify which regions show the highest concentration of inequality.

Key Objectives:

- Summarize and compare the central tendency and spread of income within each region using descriptive statistics.
- Quantify income inequality per region using the Gini coefficient and the Lorenz curve.
- Statistically test whether income differences across regions are significant, not just visually different.
- Present findings through clear visualizations that policymakers can use to prioritize interventions, directly supporting SDG 10: Reduced Inequalities.

2. Data Collection & Preparation

Dataset:

A structured income dataset such as a national household income/expenditure survey (e.g., NSSO Household Consumption Survey, World Bank LIS microdata, or an equivalent publicly available regional income dataset) is used. Each row represents one household or individual record, and columns represent the features listed below.

Feature	Description	Type
Region	State / district / urban-rural category	Categorical

Annual Income	Total reported annual household income	Numeric
Household Size	Number of members in the household	Numeric
Education Level	Highest education level of the earning member	Categorical
Occupation Type	Primary / Secondary / Tertiary sector employment	Categorical
Urban/Rural	Location classification	Categorical
Income Source	Salary, business, agriculture, remittance, etc.	Categorical

Preprocessing Steps:

- Handling missing income values — records with missing income are dropped rather than imputed, since fabricating income values would distort the very inequality measures being studied.
- Removing invalid/negative income entries and correcting obvious data-entry errors (e.g., income reported in the wrong currency unit).
- Winsorizing extreme top-end incomes (capping at the 99th percentile) to prevent a handful of very high earners from distorting the mean and standard deviation, while preserving them for percentile-based measures like the Gini coefficient, which are more robust to such outliers.
- Log-transforming income (`np.log1p`) for visualizations and normality-based tests, since raw income is typically strongly right-skewed.
- Grouping records by Region using Pandas `groupby()` to prepare per-region statistical summaries.

Sample Preprocessing Code:

```
import pandas as pd
import numpy as np

df = pd.read_csv("household_income_survey.csv")
print(df.shape)
print(df.isnull().sum())

# Drop rows with missing or invalid income
df = df.dropna(subset=['annual_income'])
df = df[df['annual_income'] > 0]

# Cap extreme outliers at the 99th percentile (winsorizing)
cap = df['annual_income'].quantile(0.99)
df['income_capped'] = df['annual_income'].clip(upper=cap)
```

```
# Log-transform for skew-robust visualization/testing
df['log_income'] = np.log1p(df['annual_income'])
```

```
# Group by region for later comparison
regions = df.groupby('region')
```

3. Analysis Using NumPy/Pandas

Pandas is used to compute per-region descriptive statistics and grouped summaries, while NumPy underlies the numerical computations for percentiles, the Gini coefficient, and array-based operations.

```
# Descriptive statistics per region
summary = df.groupby('region')['annual_income'].describe()
print(summary)
```

```
# Skewness and kurtosis per region
skew_kurt = df.groupby('region')['annual_income'].agg(['skew'])
print(skew_kurt)
```

```
# Gini coefficient function (NumPy-based)
def gini(income_array):
    x = np.sort(np.array(income_array))
    n = len(x)
    cum_x = np.cumsum(x)
    return (2 * np.sum((np.arange(1, n+1)) * x) - (n+1) * cum_x[-1]) / (n * cum_x[-1])
```

```
gini_by_region = df.groupby('region')['annual_income'].apply(gini)
print(gini_by_region.sort_values(ascending=False))
```

```
# Percentile (P90/P10) ratio per region — another common inequality measure
p90 = df.groupby('region')['annual_income'].quantile(0.90)
p10 = df.groupby('region')['annual_income'].quantile(0.10)
p90_p10_ratio = (p90 / p10).sort_values(ascending=False)
print(p90_p10_ratio)
```

Expected Insight:

This analysis typically reveals that mean income can look similar across two regions while their Gini coefficients differ substantially — meaning one region has income concentrated among a small share of high earners while the other has a more even spread, a distinction that the mean alone would completely hide. Regions with high P90/P10 ratios and high positive skewness usually correspond to the highest measured Gini coefficients, confirming that these regions carry the greatest income inequality and should be prioritized for policy attention.

4. Statistical Comparison & Hypothesis Testing

Since the goal is to compare distributions rather than build a predictive model, the 'model selection' step here is a statistical comparison framework: descriptive inequality measures (Gini coefficient, Lorenz curve) paired with formal hypothesis tests to check whether the observed regional income differences are statistically significant or could plausibly be due to random sampling variation. A one-way ANOVA is used as the primary test when income (or log-income) is reasonably close to normal within each region; the non-parametric Kruskal-Wallis test is used as a robustness check since raw income is rarely perfectly normal.

Sample Implementation Code:

```
from scipy import stats

# Prepare income arrays per region
region_groups = [group['annual_income'].values for name, group in df.groupby('region')]

# One-way ANOVA — tests whether at least one region's mean income differs significantly
f_stat, p_value_anova = stats.f_oneway(*region_groups)
print("ANOVA F-statistic:", f_stat, "p-value:", p_value_anova)

# Kruskal-Wallis — non-parametric alternative, robust to non-normal/skewed income data
h_stat, p_value_kw = stats.kruskal(*region_groups)
print("Kruskal-Wallis H-statistic:", h_stat, "p-value:", p_value_kw)

# Pairwise comparison example: Region A vs Region B (independent t-test)
region_a = df[df['region']=='Region A']['annual_income']
region_b = df[df['region']=='Region B']['annual_income']
t_stat, p_value_t = stats.ttest_ind(region_a, region_b, equal_var=False)
print("Region A vs Region B t-test p-value:", p_value_t)
```

Model/Test Evaluation:

Measure / Test	Purpose
Gini Coefficient	Single-number summary of income inequality within a region (0 = perfect equality, 1 = maximum inequality)
Lorenz Curve	Visual complement to the Gini coefficient, showing the cumulative share of income held by the poorest x% of the population
P90/P10 Ratio	Ratio of top-decile to bottom-decile income — an intuitive, outlier-resistant inequality measure
ANOVA / Kruskal-Wallis	Tests whether mean/median income differs significantly across three or more regions
Independent t-test	Tests whether two specific regions differ significantly in income

A significant ANOVA/Kruskal-Wallis result ($p < 0.05$) confirms that regional income differences are unlikely to be due to chance, justifying region-specific policy responses rather than a single nationwide approach; the Gini coefficient and P90/P10 ratio then indicate which of those regions carry the deepest structural inequality, as opposed to simply having a lower or higher average income.

5. Visualization & Interpretation

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
# Boxplot comparing income distribution across regions
sns.boxplot(x='region', y='annual_income', data=df)
plt.xticks(rotation=45)
plt.title("Income Distribution by Region")
plt.show()
```

```
# KDE plot of log-income by region — reveals shape differences hidden by the mean
sns.kdeplot(data=df, x='log_income', hue='region', common_norm=False)
plt.title("Log-Income Density by Region")
plt.show()
```

```
# Lorenz curve for a selected region
def lorenz_curve(income_array):
    x = np.sort(income_array)
    cum = np.cumsum(x) / np.sum(x)
    cum = np.insert(cum, 0, 0)
    return np.linspace(0, 1, len(cum)), cum
```

```
pop_share, income_share = lorenz_curve(df[df['region']=='Region
A']['annual_income'].values)
plt.plot(pop_share, income_share, label='Region A Lorenz Curve')
plt.plot([0,1], [0,1], linestyle='--', color='gray', label='Perfect Equality')
plt.xlabel("Cumulative share of population")
plt.ylabel("Cumulative share of income")
plt.title("Lorenz Curve — Region A")
plt.legend()
plt.show()
```

```
# Bar chart of Gini coefficients by region
gini_by_region.sort_values(ascending=False).plot(kind='bar', color='indianred')
plt.ylabel("Gini Coefficient")
plt.title("Income Inequality (Gini) by Region")
plt.show()
```

Interpretation of Visuals:

- Boxplot by Region – shows the median, interquartile spread, and outlier income earners in each region side by side, immediately highlighting which regions have both higher incomes and wider spread.
- KDE Plot (Log-Income) – reveals distribution shape differences (e.g., a long right tail in one region vs a more symmetric distribution in another) that summary statistics like the mean alone would mask.
- Lorenz Curve – the further a region's curve bows away from the diagonal 'perfect equality' line, the more unequal its income distribution — a direct, intuitive visual complement to the Gini coefficient.
- Gini Bar Chart – ranks all regions by inequality in one glance, giving policymakers a prioritized list of which regions need the most urgent intervention.

6. Innovation & SDG Mapping

This project directly aligns with United Nations Sustainable Development Goal 10 (SDG 10: Reduced Inequalities), which aims to 'reduce inequality within and among countries', including the specific target of achieving and sustaining income growth for the bottom 40% of the population at a rate higher than the national average.

Real-World Relevance:

- Evidence-Based Policy Targeting – ranking regions by Gini coefficient and P90/P10 ratio gives policymakers an objective, data-driven basis for directing welfare programs, minimum-wage policy, or tax reform toward the regions with the deepest inequality, rather than relying on average-income figures that can mask disparity.
- Monitoring Progress Over Time – repeating this same Pandas-based pipeline on successive years of survey data creates a reproducible way to track whether inequality-reduction interventions are actually working, directly supporting SDG 10's monitoring targets.
- Highlighting Structural Disparities – comparing income distributions alongside Education Level and Urban/Rural fields helps surface which structural factors (access to education, rural infrastructure) most strongly associate with regional inequality, pointing toward root-cause interventions rather than symptom management.
- Transparent, Reproducible Analysis – using open statistical measures (Gini, Lorenz curve, ANOVA) rather than a opaque model keeps the analysis auditable and explainable to non-technical policymakers and the public, which matters for public trust in equality-focused policy.

Innovation Aspect:

Rather than reporting a single national inequality figure, this project decomposes inequality region by region and pairs each Gini score with a formal significance test — turning a purely descriptive comparison into a statistically defensible ranking that distinguishes 'this region is genuinely more unequal' from 'this difference could just be sampling noise', which is exactly the level of rigor needed to justify targeted public policy.

7. Documentation & Conclusion

Summary of Workflow:

Data Collection → Preprocessing (cleaning, outlier capping, log-transform) → Exploratory Analysis (Pandas/NumPy descriptive statistics, Gini coefficient, percentile ratios) → Statistical Testing (ANOVA / Kruskal-Wallis / t-tests) → Visualization (Matplotlib/Seaborn: boxplots, KDE, Lorenz curves, Gini bar chart) → Insight Generation & SDG Mapping.

Conclusion:

The analysis demonstrates that income inequality varies meaningfully across regions in ways that simple mean-income comparisons fail to capture — two regions with similar average income can have very different Gini coefficients and Lorenz curves, reflecting fundamentally different distributions of economic opportunity. Statistical testing confirms whether these regional differences are significant rather than incidental, giving the findings enough rigor to inform real policy decisions. Beyond describing the current state, this project's real value lies in providing a reusable, transparent framework for continuously monitoring regional inequality, directly supporting SDG 10: Reduced Inequalities.

Future Scope:

- Extend the analysis to multi-year panel data to track whether regional inequality is improving or worsening over time.
- Incorporate additional dimensions of inequality (education access, healthcare spending, gender income gap) for a more holistic inequality index beyond income alone.
- Build an interactive dashboard (e.g., using Plotly/Dash) so policymakers can explore region-level Gini coefficients and Lorenz curves without needing to run the underlying code.